

Báo cáo cancellation probability — theo giá trị biến phân loại

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Mô hình: LightGBM v2 (Optuna tuned)

Phạm vi: 82,811 booking | Ngưỡng: 0.35

File kèm: data/11_cancellation_probability_scores.csv

1. Mục tiêu

Biểu mean $P(h|y)$ theo từng giá trị biến phân loại (6 biến v2), so sánh với tỉ lệ hủy thực tế và tỉ lệ dự đoán Hủy @ 0,35.

Cột bảng	Ý nghĩa
n	Số booking
mean_P_huy	Trung bình xác suất hủy dự đoán
actual_cancel_%	Tỉ lệ is_canceled=1 thực tế
predicted_cancel_%	Tỉ lệ dự đoán Hủy @ 0,35

deposit_type

3 giá trị — sắp xếp theo mean $P(h|y)$ giảm dần.

deposit_type	n	mean $P(h y)$	Hủy thực tế %	Dự đoán Hủy %
Non Refund	963	0.9533	95.0%	96.6%
No Deposit	81,767	0.3974	27.3%	50.8%
Refundable	81	0.2926	28.4%	30.9%

Top 3 rủi ro cao: Non Refund (0.953), No Deposit (0.397), Refundable (0.293)
Top 3 rủi ro thấp: Refundable (0.293), No Deposit (0.397), Non Refund (0.953)

market_segment

8 giá trị — sắp xếp theo mean $P(h|y)$ giảm dần.

market_segment	n	mean P(h _{xy})	H _{xy} th _c t _{xy} %	D _{xy} đoán H _{xy} %
Online TA	50,391	0.5032	35.5%	66.4%
Groups	3,690	0.3639	31.2%	42.5%
Undefined	2	0.3079	100.0%	0.0%
Aviation	220	0.2821	19.1%	40.0%
Direct	11,351	0.2661	14.9%	27.8%
Complementary	619	0.2425	13.1%	20.2%
Corporate	3,678	0.2351	12.8%	20.8%
Offline TA/TO	12,860	0.2049	15.1%	26.1%

Top 3 r_{xy} ro cao: Online TA (0.503), Groups (0.364), Undefined (0.308)

Top 3 r_{xy} ro th_p: Offline TA/TO (0.205), Corporate (0.235), Complementary (0.242)

country

178 giá tr_{xy} — s_{xy}p x_{xy}p theo mean P(h_{xy}) gi_{xy}m d_{xy}n.

country	n	mean P(h _{xy})	H _{xy} th _c t _{xy} %	D _{xy} đoán H _{xy} %
FJI	1	0.9493	100.0%	100.0%
NIC	1	0.8626	100.0%	100.0%
ARE	48	0.8593	85.4%	97.9%
FRO	3	0.8573	66.7%	100.0%
NAM	1	0.8370	0.0%	100.0%
BEN	3	0.8320	100.0%	100.0%
VGB	1	0.8254	100.0%	100.0%
KHM	1	0.8193	100.0%	100.0%
ATF	1	0.8158	0.0%	100.0%
IMN	2	0.8070	100.0%	100.0%
HND	1	0.7988	100.0%	100.0%

KIR	1	0.7916	0.0%	100.0%
GLP	1	0.7386	100.0%	100.0%
SAU	45	0.7235	66.7%	86.7%
JEY	8	0.7055	100.0%	87.5%
BGD	9	0.7020	77.8%	100.0%
TJK	2	0.6931	50.0%	100.0%
MYT	2	0.6611	100.0%	100.0%
BHR	5	0.6469	80.0%	100.0%
HKG	23	0.6468	91.3%	91.3%
RWA	2	0.6458	0.0%	100.0%
MRT	1	0.6390	0.0%	100.0%
AGO	339	0.6290	56.3%	66.1%
PAN	8	0.6231	0.0%	75.0%
GHA	4	0.6206	50.0%	100.0%
MAC	10	0.6196	90.0%	90.0%
GEO	19	0.6090	73.7%	73.7%
CHN	785	0.5938	49.4%	72.0%
PRY	4	0.5848	0.0%	75.0%
SEN	11	0.5838	72.7%	72.7%
QAT	14	0.5826	71.4%	78.6%
MAR	229	0.5482	45.4%	65.9%
PHL	31	0.5438	61.3%	74.2%
TUR	201	0.5419	47.8%	67.7%
SLE	1	0.5390	0.0%	100.0%
AND	7	0.5373	71.4%	85.7%
PAK	13	0.5274	61.5%	69.2%

RUS	545	0.5184	39.4%	65.5%
LUX	253	0.5161	39.1%	71.1%
TZA	5	0.5119	60.0%	100.0%
PRT	25,299	0.5072	36.8%	62.1%
AZE	16	0.5060	50.0%	68.8%
NCL	1	0.5044	0.0%	100.0%
MDV	12	0.4996	75.0%	58.3%
BRA	1,936	0.4965	36.8%	65.8%
KOR	118	0.4922	36.4%	66.1%
ITA	2,900	0.4910	36.1%	63.4%
KWT	16	0.4897	37.5%	62.5%
ABW	2	0.4893	0.0%	50.0%
IDN	25	0.4807	56.0%	56.0%
ZMB	2	0.4789	50.0%	100.0%
ASM	1	0.4776	0.0%	100.0%
TWN	41	0.4749	31.7%	65.9%
GIB	14	0.4726	78.6%	78.6%
ZAF	78	0.4703	37.2%	57.7%
HUN	199	0.4698	32.2%	63.3%
DOM	12	0.4678	50.0%	58.3%
GGY	2	0.4626	100.0%	50.0%
TMP	3	0.4581	33.3%	66.7%
UMI	1	0.4566	100.0%	100.0%
ISR	370	0.4541	35.4%	63.8%
CPV	23	0.4524	52.2%	60.9%
ETH	3	0.4432	33.3%	66.7%

MLT	17	0.4398	29.4%	58.8%
THA	38	0.4390	36.8%	55.3%
SGP	34	0.4386	35.3%	61.8%
URY	28	0.4342	32.1%	71.4%
DMA	1	0.4332	0.0%	100.0%
NOR	491	0.4324	28.7%	54.8%
SVK	48	0.4321	20.8%	56.2%
TUN	37	0.4273	48.6%	62.2%
MUS	7	0.4244	14.3%	71.4%
UGA	2	0.4198	0.0%	50.0%
PLW	1	0.4196	0.0%	100.0%
BLR	24	0.4191	33.3%	54.2%
COM	2	0.4179	0.0%	50.0%
BIH	11	0.4114	27.3%	54.5%
IRN	77	0.4083	27.3%	61.0%
ARG	199	0.4037	25.6%	53.8%
SLV	2	0.3999	0.0%	50.0%
EST	77	0.3987	23.4%	51.9%
COL	68	0.3978	33.8%	55.9%
CHE	1,545	0.3955	24.8%	54.8%
CZE	128	0.3952	25.8%	53.1%
UKR	60	0.3895	33.3%	51.7%
ROU	447	0.3891	26.2%	51.7%
HRV	91	0.3883	25.3%	53.8%
NGA	30	0.3875	56.7%	50.0%
IND	138	0.3869	23.9%	51.4%

GAB	4	0.3867	50.0%	50.0%
EGY	30	0.3857	33.3%	56.7%
AUS	370	0.3846	24.6%	50.8%
ESP	6,973	0.3829	26.2%	50.2%
CIV	6	0.3822	33.3%	50.0%
USA	1,825	0.3799	24.8%	50.3%
GRC	114	0.3769	27.2%	50.0%
VEN	21	0.3759	38.1%	47.6%
MNE	5	0.3750	40.0%	40.0%
LIE	3	0.3743	33.3%	66.7%
ARM	8	0.3702	25.0%	50.0%
DNK	373	0.3700	24.4%	46.1%
ISL	22	0.3687	18.2%	50.0%
KAZ	19	0.3657	26.3%	47.4%
MKD	3	0.3652	33.3%	33.3%
UZB	4	0.3622	50.0%	50.0%
VNM	8	0.3618	25.0%	50.0%
KEN	6	0.3611	33.3%	16.7%
MCO	4	0.3568	25.0%	50.0%
SVN	44	0.3561	20.5%	43.2%
SUR	5	0.3509	0.0%	40.0%
MMR	1	0.3504	0.0%	100.0%
IRL	2,893	0.3502	22.2%	46.6%
MOZ	50	0.3447	32.0%	42.0%
SWE	783	0.3391	22.2%	42.3%
CMR	10	0.3381	0.0%	50.0%

ZWE	4	0.3380	50.0%	50.0%
CYP	40	0.3374	20.0%	45.0%
DZA	78	0.3365	26.9%	41.0%
CHL	63	0.3293	22.2%	41.3%
POL	739	0.3280	22.1%	42.4%
LBN	28	0.3229	21.4%	42.9%
BEL	2,022	0.3229	20.2%	38.0%
FRA	8,464	0.3188	20.1%	41.9%
CN	1,055	0.3180	21.1%	42.0%
ECU	24	0.3160	29.2%	41.7%
LVA	47	0.3155	17.0%	42.6%
DEU	5,109	0.3135	20.2%	41.8%
SYC	2	0.3127	50.0%	50.0%
NLD	1,864	0.3092	18.6%	37.6%
BGR	67	0.3067	13.4%	40.3%
GBR	9,921	0.2970	19.6%	39.7%
MYS	24	0.2947	12.5%	41.7%
AUT	887	0.2936	17.9%	36.3%
JOR	18	0.2912	16.7%	27.8%
ATA	2	0.2898	0.0%	50.0%
CUB	8	0.2846	0.0%	37.5%
MEX	71	0.2830	12.7%	33.8%
CRI	18	0.2808	5.6%	33.3%
FIN	416	0.2686	15.4%	27.4%
PER	25	0.2637	24.0%	32.0%
ALB	11	0.2602	18.2%	36.4%

JPN	176	0.2514	13.6%	23.3%
OMN	18	0.2453	22.2%	33.3%
SRB	47	0.2309	6.4%	34.0%
GTM	3	0.2041	0.0%	33.3%
TGO	2	0.2001	0.0%	50.0%
BOL	3	0.1997	0.0%	0.0%
LTU	73	0.1980	9.6%	21.9%
BFA	1	0.1926	0.0%	0.0%
LKA	5	0.1899	0.0%	20.0%
NZL	63	0.1874	6.3%	20.6%
JAM	6	0.1854	0.0%	33.3%
BWA	1	0.1647	0.0%	0.0%
LAO	2	0.1636	0.0%	0.0%
IRQ	10	0.1484	0.0%	20.0%
GNB	9	0.1472	11.1%	11.1%
SYR	3	0.1441	0.0%	33.3%
AIA	1	0.1372	0.0%	0.0%
Unknown	437	0.1354	7.6%	9.4%
PRI	11	0.1079	9.1%	9.1%
MLI	1	0.0973	0.0%	0.0%
CAF	5	0.0929	0.0%	0.0%
DJI	1	0.0853	0.0%	0.0%
NPL	1	0.0761	0.0%	0.0%
SMR	1	0.0685	0.0%	0.0%
STP	2	0.0670	0.0%	0.0%
CYM	1	0.0642	0.0%	0.0%

LBY	7	0.0634	0.0%	0.0%
MWI	2	0.0528	0.0%	0.0%
KNA	2	0.0398	0.0%	0.0%
PYF	1	0.0372	0.0%	0.0%
MDG	1	0.0257	0.0%	0.0%
GUY	1	0.0257	0.0%	0.0%
SDN	1	0.0253	0.0%	0.0%
LCA	1	0.0252	0.0%	0.0%
BRB	1	0.0226	0.0%	0.0%
BDI	1	0.0224	0.0%	0.0%
BHS	1	0.0104	0.0%	0.0%

Top 3 r*■*i ro cao: FJI (0.949), NIC (0.863), ARE (0.859)
Top 3 r*■*i ro th*■*p: BHS (0.010), BDI (0.022), BRB (0.023)

distribution_channel

5 giá tr*■* — s*■*p x*■*p theo mean $P(h\text{■}y)$ gi*■*m d*■*n.

distribution_channel	n	mean $P(h\text{■}y)$	H <i>■</i> y th <i>■</i> c t <i>■</i> %	D <i>■</i> <i>■</i> oán H <i>■</i> y %
TA/TO	65,956	0.4408	31.5%	57.7%
Undefined	5	0.3804	80.0%	60.0%
GDS	172	0.3345	19.8%	44.2%
Direct	12,291	0.2663	15.1%	27.6%
Corporate	4,387	0.2343	13.6%	22.2%

Top 3 r*■*i ro cao: TA/TO (0.441), Undefined (0.380), GDS (0.335)
Top 3 r*■*i ro th*■*p: Corporate (0.234), Direct (0.266), GDS (0.335)

customer_type

4 giá tr*■* — s*■*p x*■*p theo mean $P(h\text{■}y)$ gi*■*m d*■*n.

customer_type	n	mean P(h _h y)	H _h y th _c t _h %	D _h o _h án H _h y %
Transient	69,939	0.4369	30.4%	56.1%
Transient-Party	9,294	0.2339	15.8%	26.9%
Contract	3,068	0.2031	16.5%	23.8%
Group	510	0.1554	10.6%	14.3%

Top 3 r_hi ro cao: Transient (0.437), Transient-Party (0.234), Contract (0.203)

Top 3 r_hi ro th_hp: Group (0.155), Contract (0.203), Transient-Party (0.234)

hotel

2 giá tr_h — s_hp x_hp theo mean P(h_hy) gi_hm d_hn.

hotel	n	mean P(h _h y)	H _h y th _c t _h %	D _h o _h án H _h y %
City Hotel	50,686	0.4383	30.7%	57.3%
Resort Hotel	32,125	0.3491	24.1%	42.0%

Top 3 r_hi ro cao: City Hotel (0.438), Resort Hotel (0.349)

Top 3 r_hi ro th_hp: Resort Hotel (0.349), City Hotel (0.438)

Ghi chú

- Mean P(h_hy) theo nhóm không **nh_hng** ngh_ha xác su_ht nhân qu_h — ch_h ph_hn ánh scoring trung bình c_ha mô hình trên segment **h_h**.
- Bi_hn country có nhi_hu m_hc hi_hm; các n_hc ít booking có mean P(h_hy) bi_hn **nh_hng** m_hnh.
- So sánh actual_cancel_% vs mean_P_huy giúp **nh_h**ánh giá calibration theo segment.

Sinh t_h **nh_hng** b_hi _run_11_probability_reports.py.